



AHMANE ZOUBIDA

Born on May 23rd, 1990
Batna – Algeria

✉ zoubida.ahmane@univ-batna.dz
✉ ahmane_zoubida@yahoo.fr

Scientific interests

- Theoretical star formation :
Accretion disk, Jets and Outflow.
- Computational fluid Dynamics:
HD, MHD and Ambipolar
diffusion.
- Inter-Stellar Medium :
Shock-Cloud Interaction, Collapse
of molecular cores.

Education

2020. PhD in Astrophysics & radiation physics

Title: Stellar simulation
using PLUTO code.

Supervisor: A. Bouldjedri.

Co-Supervisor: S. Massaglia.
Batna 1 University, Algeria.
Universita degli studi di
torino, Italy.

2013. Masters in Matter and radiation Physics

Batna 1, University, Algeria

2010. Bachelor in Physics

Sciences of Matter Faculty

Current position

2020 – Present. Faculty member at Batna 1 University in
Algeria, specifically in the Faculty of Matter Sciences,
Physics Department.

- Astrophysical fluid dynamics.
- Numerical analysis .
- Plasma astrophysics.

Publications

Ahmane, Z., Mignone, A., Zanni, C. et al. (2020) Simulations of
protostar-driven photoionization in Herbig-Haro jets. *Astrophys
Space Sci* 365, 94.
doi: <https://doi.org/10.1007/s10509-020-03808-3>

Oral communications

18/29 August 2014 : College on advanced plasma physics:
Accretion disc luminosity in different models , Poster
presentation. Trieste, Italy university.

24/25 May 2015 : 2eme journ  e d'  tude des jeune
scientifique :Jets around young stellar objects YSO , Poster
presentation Batna university.

6 May 2015, Jets launching and propagation simulation. oral
presentation, Batna, Algeria.

23 May /1 June 2016, International astrophysical jet school:
Simulation of photoionisation effect into YSO, poster
presentation, Corsica, France .

March 2017, Incorporating ambipolar diffusion into PLUTO code,
poster presentation, Khnechela, Algeria.

Numerical tools

Python,C++,C ● ● ● ●
HTML,Css ● ● ● ●
Mathematica ● ● ● ●
Linux ● ● ● ●
LaTex ● ● ● ●
PLUTO code ● ● ● ●

Languages

Arabic ● ● ● ●
French ● ● ● ●
English ● ● ● ●
Italian ● ● ● ●

Professional Referee

Professor Silvano Massaglia

Department of Physics, University of Turin.
silvano.massaglia@unito.it

Professor Abdelhamid Bouldjedri

Department of Physics, University of Batna1.